

REPUBLIC OF RWANDA



**MINISTRY OF EDUCATION
EASTERN PROVINCE
KIREHE DISTRICT**

Module Code: SWDFA501
Module Name: front end
application with react js

Date: 12/03/2026

Period: 08H30 – 11H30 AM

END OF TERM II EXAMINATIONS

QUESTION PAPER

SECTOR: ICT AND MULTIMEDIA

RTQF LEVEL: 3

TRADES: SOFTWARE DEVELOPMENT

MARKS:

/100

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Three Sections (A, B, C). Follow the instructions given below, and answer the indicated questions for a total of 100marks

Section A: Seventeen (17) questions, all Compulsory 55 marks

Section B: Among the four(4) questions, attempt any three (3)30 marks

Section C: Among the two (2) questions, attempt any one (1)15 marks

Section A: Seventeen (17) questions, all Compulsory**55 marks**

1. Match the react concepts in column A with their description in column B and write the corresponding answers in the space provided **/5marks**

Column A	Column B	Answer
1.component	A. A special kind of object that allow developers to write HTML-like syntax in java script	1= C
2.jsx	B. Java script features that allow components to interact with lifecycle event or manage	2=A
3.redux	C.A reusable function or class that renders a UI element and its behavior	3=D
4.React Rout	D. A tool used for managing global state in an application	4=E
5.Hooks	E.A way to handle navigation and dynamic routing within a react application	5=B

2. Read carefully the following statements about Tailwind CSS framework and answer by TRUE if the statement is correct and by FALSE if the statement is incorrect. **/5marks**

- a) Tailwind CSS does not support grid-based layouts for creating flexible, responsive designs. **FALSE**
- b) The mobile-first approach means designing for smaller screens first and progressively enhancing the layout for larger screens. **TRUE**
- c) Tailwind CSS does not allow developers to customize typography or scale font sizes for responsive design. **FALSE**
- d) Tailwind CSS offers utility classes to style interactive elements such as buttons, forms, and hover states, ensuring responsiveness. **TRUE**
- e) Testing responsiveness and iterating on design are not necessary when using Tailwind CSS because it automatically makes everything responsive. **FALSE**

3. Which CSS property is used to make text bold? **/3marks**
- a) font-family
 - b) font-weight**
 - c) font-size
 - d) font-style
4. Read the flowing sub-questions and circle the corresponding to the right option to each questions the given **/3marks**
- I. Which of the following best describes 'props' in React?
- a) Mutable data stored inside a component
 - b) Built-in lifecycle methods
 - c) A way to fetch data from an API
 - d) Read-only attributes passed from parent to child components**
- II. Which command is used to install React Router in a project?
- a) npm install react-route
 - b) npm add router
 - c) npm install react-router-dom**
 - d) npx react-router install
5. Match each term in Column A with its correct definition in Column B. **/5marks**

Answers	Column A	Definition (Column B)
1=E	useState	A. A syntax extension allowing HTML-like code to be written in JavaScript files
2= D	Component WillUnmount	B. A library for routing in React, enabling navigation between views
3= B	React Router	C. A predictable state container for managing global application state
4= A	JSX	D. Called just before a component is removed from the DOM

5= C	Redux	E. A hook that allows functional components to have and update state
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6. Which of the following is TRUE about functional components compared to class components? **/3marks**

- a) Functional components cannot manage state
- b) Class components are preferred in modern React
- c) Functional components are more concise and use hooks**
- d) Functional components require a render() method

7. Match the following terms with their corresponding statements in the table below. **/4marks**

ANSWERS	Term	Explanation
1= C	1. components	A. Are used to define the mapping between URLs (or paths) and the Vue components that should be displayed when those URLs are visited.
2= A	2. routes	B. Refers to a specific URL on a server that is used to interact with a backend API.
3= D	3. State management	C. They can be reused throughout your application. Once defined, you can use them multiple times in different parts of your app.
4=B	4. API Endpoint	D. to the management of application state across components.

8. Which tool requires creating reducers and a store? **/3marks**

- a) Context API

- b) Redux**
 - c) useRef
 - d) useMemo
9. What is the first step in implementing useState? /3marks
- a) Define reducer
 - b) Import useState from React**
 - c) Create context
 - d) Install Redux
10. Which HTML tag is most important for SEO page titles? /3marks
- a) <meta>
 - b) <div>
 - c) <title>**
 - d) <header>
11. Which tool can you use to test website performance? /3marks
- a) Microsoft Word
 - b) Google PageSpeed Insights**
 - c) Notepad
 - d) MySQL
12. Structured data format commonly used in SEO is: /3marks
- a) XML
 - b) JSON-LD**
 - c) CSV
 - d) TXT
13. What is the main purpose of robots.txt? /3marks
- a) Store user data
 - b) Block hackers
 - c) Tell search engines which pages to crawl**
 - d) Improve CSS design
14. Which command creates a new Next.js project? /3marks
- a) npm install next
 - b) npx create-next-app@latest**

- c) node next start
 - d) npm run build
15. To install Tailwind CSS in Next.js, which command is correct? **/2marks**
- a) npm install tailwindcss**
 - b) npm create tailwind
 - c) next install tailwind
 - d) node install tailwind
16. Which of the following is correct when importing Google Fonts using next/font? **/2marks**
- a) import { Roboto } from 'next/font/google'**
 - b) import Roboto from 'google/font'
 - c) import font from 'next/google'
 - d) import googleFont from 'next'
17. What is SEO an abbreviation for? **/2marks**
- a) Site Engine Optimization
 - b) Search Engine Optimization**
 - c) Search Engine Operations
 - d) Site Evaluation Online

Section B: Among the five (5) questions, attempt any three (3)30 marks

18. determine whether the following statement are True or False **/10 marks**
- a) Next.js automatically caches fetch () requests in Server Components by default. **True**
 - b) Request Memorization stores data permanently in the database. **False**
 - c) Time-based revalidation allows cached data to update after a specific time interval. **True**
 - d) On-demand revalidation allows developers to manually revalidate cached pages. **True**

e) Sensitive user-specific data should always be cached for better performance. **False**

19. Emmy is a front end developer working a browser-based user interface (UI) using react.js at update tech LTD company. He is exploring different style approaches and is considering using SASS (syntactically awesome style sheets) for managing the user interface (UI) visual design. Before deciding, she wants to weigh the pros and cons of using SASS specifically for styling user interface environments. Basing Emmy scenario discuss five (5) advantages and five (5) disadvantage of using SASS for styling user interface environment in a react.js project.
/10marks

Answers

Advantages of Using SASS in React.js Projects

1. **Nesting Support:** SASS allows nesting of CSS selectors, making the styles more organized and readable, which is helpful for complex React components.
2. **Variables:** You can define variables for colors, fonts, spacing, etc., ensuring consistency across the UI and making theme changes easier.
3. **Mixins & Functions:** Reusable chunks of code (mixins) and functions allow Emmy to avoid repeating style patterns, reducing redundancy.
4. **Partials and Imports:** SASS lets you split styles into multiple files (partials) and import them, improving maintainability in large React projects.
5. **Mathematical Operations & Logic:** SASS supports calculations and conditionals in styles, enabling dynamic adjustments like responsive spacing or color manipulation.

Disadvantages of Using SASS in React.js Projects

1. **Build Process Dependency:** SASS requires compilation to CSS, adding an extra step in the build process and potentially increasing build time.
2. **Learning Curve:** Developers need to learn SASS syntax and features, which may slow down onboarding for new team members.
3. **Over-Nesting Risk:** Excessive nesting can lead to very specific selectors that are hard to override, causing maintainability issues.

4. **Global Scope Issues:** By default, SASS generates global CSS, which may cause style conflicts across React components unless additional scoping solutions (like CSS Modules) are used.
5. **Debugging Complexity:** Compiled CSS can be harder to debug compared to plain CSS, especially when using nested selectors, mixins, or imports.

20. Matching the following command installation library and functions

/10marks

Command	Function	Answer
1. npm install axios	A. Used to create navigation between pages in a React application	1=C
2. npm install tailwindcss	B. Used to manage global state in large applications	2=D
3. npm install redux	C. Used to make HTTP requests to APIs	3=B
4. npm install react-router-dom	D. Used for styling using utility-first CSS classes	4=A
5. npm install react-icons	E. Used to add ready-made icons to React components	5=E

21. Your development team is building a responsive web application using Tailwind CSS. The project primarily targets specific device widths: iPhone SE (375px) iPad (810px) Desktop (1440px). During planning, the team is debating whether to: Use Tailwind's default responsive breakpoints (sm, md, lg, xl, 2xl), or Define custom breakpoints in the tailwind.config.js file to exactly match the specified device widths. Critically evaluate both approaches in terms of: Scalability, Maintainability, Flexibility, Responsive design best practices. Based on your analysis, recommend the most appropriate solution for the project and justify your answer with clear reasoning. **/10marks**

Evaluation:

Default breakpoints (sm, md, lg, xl, 2xl):

Scalability: High – works across many devices.

Maintainability: Easy – standard and familiar.

Flexibility: High – mobile-first, fluid design.

Best practices: Aligns with responsive design principles.

Drawback: Minor mismatch with exact device widths (375px, 810px), but manageable.

Custom breakpoints (375px, 810px, 1440px):

Scalability: Low – tied to specific devices.

Maintainability: Harder – custom names and widths.

Flexibility: Low – rigid, less adaptable.

Best practices: Device-specific, not content-first.

Benefit: Exact control for target devices.

22. A developer created the following code for a responsive heading:

```
<h1 className="text-4xl text-4xl md:text-4xl lg:text-5xl font-  
bold">
```

```
Welcome
```

```
</h1> Analyze this code and identify the redundancy issue.  
Explain why this is inefficient and propose a better solution  
following the mobile-first approach / 10marks
```

Answers

Redundancy Issue

- **Redundant Class Usage:** The class `text-4xl` is specified three times: once for the base styles, once for the `md` breakpoint, and again for the `lg` breakpoint.
- **Efficiency:** This redundancy leads to an unnecessarily large class list. Each time a class is repeated, it creates clutter in the code, making it harder to read and maintain.

Why this is Inefficient

Increased Code Size: Redundant classes balloon the size of the HTML and can lead to longer render times since more classes need to be processed.

Maintainability: If a developer needs to change the size across various breakpoints, they would need to update multiple instances of `text-4xl`, increasing the potential for errors.

Clarity: It can confuse other developers who read the code, making it less clear which class should take priority.

Section C: Among the two (2) questions, attempt any one (1) 15 marks

23. Analyze the following picture and answer bellow questions related to it.



a) Differentiate router, routes and route / **3marks**

Router: A component that provides routing context for a React application, enabling navigation between pages.

Routes: A container that groups multiple Route components and renders the first one that matches the URL.

Route: A component that maps a specific URL path to a React component to be displayed.

b) As you see this picture what is the router view? **/3marks**

COUNT

c) Which event handler is used when clicking the "incrising" button?

/3marks

i. onChange

ii. onClick

iii. onSubmit

iv. onHover

d) Write a complete functional component for the Counter page shown in the image. **/7marks**

```
import React, { useState } from 'react';
const Counter = () => {
  const [count, setCount] = useState(11);
  const increment = () => {
    setCount(count + 1);
  };
  const decrement = () => {
    setCount(count - 1);
  };

  return (
    <div style={{ textAlign: 'center', marginTop: '50px' }}>
      <h1>count: {count}</h1>
      <button onClick={increment}>incrising</button>
      <button onClick={decrement}>decremnt</button>
    </div>
  );
};
export default Counter;
```

24. You are working as a junior React developer at ABC Shopping Ltd, a company building an online shopping dashboard for customers in Rwanda. Your task is to improve the product search and interaction section of the dashboard. /10marks

Users should be able to:

- ✓ Search for products by typing in a search bar.

- ✓ Click a product to view its details.
- ✓ Double-click a product to add it to favorites.
- ✓ Hover over a product to highlight it.
- ✓ Prevent too many API requests when the user types very fast in the search bar.
- ✓ Ensure certain functions do not run repeatedly when the user scrolls quickly.

The senior developer tells you to use React Synthetic Events, understand event bubbling, and apply debouncing and throttling for performance

- a) What do you understand by Events in React based on the scenario? **/3marks**

Events in React are actions triggered by user interactions (like clicks, typing, or hovering) or browser actions (like scrolling).

- b) Identify and explain any three types of events used in the scenario above. **/3marks**

- i. **Click:** Triggered when the user clicks on an element.
- ii. **Keyboard:** Triggered when the user types on the keyboard.
- iii. **Mouse:** Triggered by mouse interactions like hover, move, or scroll.

- c) Describe what Synthetic Events are and explain why React uses them instead of native DOM events. **/3marks**

Synthetic Events are React's cross-browser wrapper around native DOM events.

Reason for use:

1. Ensures **consistent behavior** across different browsers.
2. Improves **performance** by using event delegation (events are handled at the root instead of attaching many listeners).
3. Allows React to **pool and reuse events** for efficiency.

- d) Explain Event Bubbling and describe one situation in the scenario where it may occur. **/3marks**

Event bubbling is when an event triggered on a child element propagates (bubbles) up to its parent elements unless stopped.

- e) Explain the terms Debouncing and Throttling, and show how each would help the search bar or scrolling actions in the scenario.

/3marks

Debouncing: A technique that delays the execution of a function until after a certain period of inactivity, preventing it from running too frequently.

Throttling: A technique that ensures a function is executed at most once in a specified time interval, controlling its rate of execution.